

Marco Fernstaedt

Software Developer

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GitHub: github.com/MarcoFernstaedt

Portfolio: marcofernstaedt.com

SUMMARY

Software developer building production grade full stack web applications with React, Next.js, TypeScript, Node.js, Express, and MongoDB. Hands on experience delivering authentication systems, real time features, AI API integrations, accessibility focused interfaces, and complete deployment workflows on Vercel and Linux. Customer service background supports user focused debugging, clear written communication, and practical product thinking.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, HTML, CSS, Python

Frontend: React, Next.js, Vite, Tailwind CSS, Framer Motion, Zustand, Monaco Editor, responsive design

Backend: Node.js, Express, REST APIs, WebSockets, Socket.IO, JWT authentication, Clerk

Databases: MongoDB, Mongoose

APIs and Services: OpenAI APIs, Stream Video, Stream Chat, Piston API, Resend, Cloudinary, Arcjet

DevOps and Tools: Git, GitHub, Vercel, Linux, DNS, SSL, npm, Chrome DevTools

Practices: Accessibility (WCAG, ARIA, screen reader testing), SEO, Schema.org, debugging, documentation, deployment verification

PROFESSIONAL EXPERIENCE

Independent Software Developer | Phoenix, AZ | 2025 to Present

Build and deploy production full stack web applications across Next.js, React, TypeScript, Node.js, Express, MongoDB, OpenAI APIs, Resend, Vercel, and Linux based development environments.

Build and maintain a public developer portfolio using Next.js, TypeScript, Tailwind CSS, Framer Motion, Vercel Analytics, GitHub metadata APIs, structured project data, and responsive project modals.

Author developer facing documentation, proof notes, deployment checks, and repeatable verification steps across portfolio, client, and application work.

Use Git, GitHub, browser debugging, API testing, build verification, and production deployment checks to ship and validate changes across multiple projects.

Saguaro Blossoms Learning Services | Web Developer (Contract) | 2025

Live: <https://saguaroblossomslearningservices.com> | GitHub: <https://github.com/MarcoFernstaedt/cynthia-tutoring-platform>

Built a production Next.js, TypeScript, and Tailwind CSS client website for a tutoring business with responsive pages, mobile navigation, bilingual English and Spanish content, pricing sections, and service conversion flows.

Implemented a server side contact API using the Next.js Node runtime and Resend, including required field validation, email validation, honeypot spam prevention, HTML escaping, owner notification emails, and visitor confirmation emails.

Delivered SEO infrastructure with canonical metadata, Open Graph and Twitter previews, sitemap and robots generation, service specific landing pages, and Schema.org data for organization, local business, services, person, and FAQ entities.

Added Vercel Analytics event tracking for language switches, consultation clicks, service clicks, pricing views, navigation, SEO topic clicks, and contact form start and submit events.

Configured custom domain, DNS, SSL, and Vercel production deployment with post launch verification checks.

PROJECTS

AI Image to Audio Accessibility Tool | Next.js, TypeScript, OpenAI APIs

GitHub: https://github.com/MarcoFernstaedt/image_accessibility_tool | Live: <https://ita-orpin.vercel.app>

Built a Next.js and TypeScript image accessibility platform that converts uploaded images into concise spoken descriptions for blind and low vision users.

Implemented an OpenAI Vision to text to speech pipeline with image analysis, generated descriptions, MP3 audio responses, and browser Blob playback and download flows.

Engineered secure image ingestion using multipart FormData parsing, MIME validation, 10 MB file limits, base64 Data URL conversion, and Node.js Buffer based audio streaming.

Added accessibility and production controls including semantic labels, ARIA live status messaging, keyboard friendly UI, Arcjet shield rules, bot detection, hosting IP checks, and token bucket rate limiting.

Real Time Messaging Platform | React, TypeScript, Node.js, Express, MongoDB, Socket.IO

GitHub: https://github.com/MarcoFernstaedt/socketio_chat_app

Developed a full stack real time chat application with React, TypeScript, Node.js, Express, MongoDB, Mongoose, and Socket.IO for authenticated one to one messaging.

Implemented secure authentication with bcrypt password hashing, JWT HTTP only cookies, protected REST endpoints, safe user serialization, and authenticated WebSocket handshakes.

Built real time messaging infrastructure with online presence broadcasts, direct message socket delivery, connection and disconnection handling, and MongoDB backed conversation history.

Created a responsive chat frontend using Zustand state management, optimistic UI updates, image message support through Cloudinary, notification sounds, and contacts and chats filtering.

InterviewLab Code Live Platform | React, TypeScript, Express, MongoDB, Clerk, Stream Video, Stream Chat, Monaco Editor

GitHub: https://github.com/MarcoFernstaedt/code_live_platform

Built a live coding interview platform with React, TypeScript, Express, MongoDB, Clerk authentication, Stream Video, Stream Chat, and Monaco Editor.

Designed protected session APIs for creating, joining, listing, and ending coding sessions with host and participant authorization, active and completed lifecycle states, and MongoDB persistence.

Integrated Stream Video and Stream Chat so each coding session provisions a dedicated video call and chat channel, adds participants on join, and tears down resources when the host ends the session.

Implemented an in browser coding workspace with Monaco Editor, resizable panels, JavaScript, Python, and Java support, and Piston API code execution with compile error, runtime error, and timeout handling.

EDUCATION AND CERTIFICATIONS

CompTIA Network Plus (In Progress)

Self directed software development training through production style projects in React, TypeScript, Next.js, Node.js, MongoDB, authentication, APIs, real time systems, AI integrations, accessibility, and deployment workflows.